Water Quality C - Water Quality - Mililani Div C Invitational - 01-22-2021

Each section of the test is labeled and is worth one-third of your score. There are 120 questions and each question is worth 1 point. Questions in which you are asked to select all corre answers may only have one correct answer, or they may have multiple. You need to select all of the correct answers and no incorrect answers to earn the point for that question.

Tiebreaking procedure:

1. The first tiebreaker is Question 13
2. The second tiebreaker is total score on Part 1
3. The third tiebreaker is total score on Part 3
4. The fourth tiebreaker is total score on Part 2
5. The fifth tiebreaker is time spent on the exam

Part 2: Coral Reef Macroflora and Fauna Identification

(I know it's out of order; it was too tedious to fix)

Refer to this image for Questions 1-4



1. (1.00 pts) What is the common name of this organism? (lowercase)

Expected Answer: banded coral shrimp

2. (1.00 pts) The term that best describes the reproductive habits of this organism is

- A) polygamy
- B) monogamy
- C) asexual

D) promiscuity

3. (1.00 pts) What is the primary diet of this organism?

A) phytoplankton from filter feeding
B) detritus that has settled
C) parasites living on fish
D) algae

4. (1.00 pts) What type of habitat does this organism live in?

A) Coral reefs
B) Freshwater estuaries
C) Oceans
D) Intertidal zones

Use this image for Questions 5-8



5. (1.00 pts) What is the common name of this organism? (lowercase)

Expected Answer: triton

6. (1.00 pts) What best describes the reproductive habits of this organism?

A) sexual reproduction with external fertilization
B) sexual reproduction with internal fertilization
C) asexual reproduction by propagation
D) asexual reproduction by parthenogenesis

7. (1.00 pts)
This is an image of this organism eating a crown of thorns starfish. The crown of thorns starfish is known for being covered in venomous spikes. What best describes this organism's ability not to be harmed by the spikes?



- A) It is entirely immune to the venom
- B) It eats the starfish from the underside, so it never comes into contact with the spikes
- C) It cuts off the spikes with its toothy radula
- D) it spits out the venemous spikes after ingesting them

8. (1.00 pts)
The crown of thorns starfish can cause destruction to large parts of coral reefs. Because this organism is one of very few natural predators of the starfish, this organism could be known as a

- A) prey animal
- B) omnivore
- C) keystone species
- D) apex producer

Refer to this image for questions 9-11



9. (1.00 pts) What is the common name of this organism? (lowercase)

Expected Answer: butterfly fish

10. (1.00 pts) What is the primary diet of this organism?

- (●) A) coral reefs
- ( ) B) krill
- ( ) C) plankton
- ( ) D) eggs of other fish

11. (1.00 pts) What best describes the mating habits of this organism?

- ( ) A) The female produces an adhesive to stick the eggs to rocks and defends them
- (●) B) the eggs float with the current until hatching
- ( ) C) the fish gives live birth
- ( ) D) the male protects the fertilizes eggs in his mouth until they hatch

12. (1.00 pts) Identify this fish, whose juveniles swim in a pattern that may mimic certain parasites so that they can avoid predation. (lowercase)



Expected Answer: sweetlips

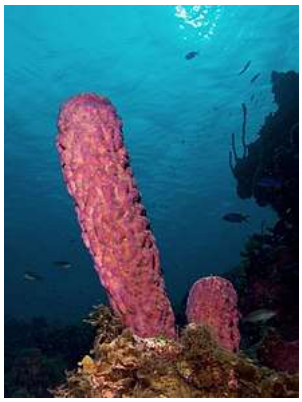
13. (1.00 pts)
Describe the role of fleshy (coralline) algae in a coral reef ecosystem. In your response, mention the common name of at least one other organism. (tiebreaker question)

Expected Answer: food source for sea urchin, parrot fish, limpets, chitons

14. (1.00 pts) Fleshy (corraline) algae may grow a thallus composed of this chemical

Expected Answer: calcium carbonate

The field of science that studies the phylum Porifera is known as spongiology. Below is an example of one such member of Porifera. It is known as the stove-pipe sponge. Refer to this image for questions 15-16



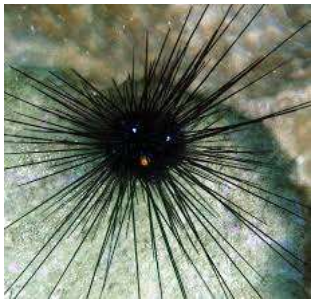
15. (1.00 pts) Organisms such as this one consume their food are known as

- (●) A) filter feeders
- ( ) B) obligate predators
- ( ) C) leukocytes
- ( ) D) decomposers

16. (1.00 pts) The process by which food enters the cells of these organisms packaged in vesicles is known as

- ( ) A) osmosis
- ( ) B) diffusion
- ( ) C) vessiclization
- (●) D) phagocytosis

Refer to this image for questions 17-20



17. (1.00 pts) Identify the common name of this organism

Expected Answer: long-spined black sea urchin

18. (1.00 pts) Describe the ecological importance of this organism to coral reefs

Expected Answer: eat algae that could overgrow and smother the reef

19. (1.00 pts) Low population density is a challenge to the reproduction of this species because

- A) they are blind and cannot find each other for internal fertilization
- B) their chemical signals become so dilute and cannot find each other for internal fertilization
- C) females only release several eggs at a time
- D) the gametes become so dilute that they may never successfully fertilize externally

20. (1.00 pts) Global climate change is a risk to the reproduction of this species because

- A) powerful storms will dislodge them from their habitats
- B) the sex determination of this species is temperature-dependent and warmer waters will only produce males
- C) the protein-shell on the eggs will denature and the eggs will not develop
- D) the calcium carbonate shell protecting the male's penis will dissolve in more acidic waters, leading to a higher risk of injury

Refer to this image for questions 21-23



21. (1.00 pts) Identify the common name of this organism

Expected Answer: barramundi cod

22. (1.00 pts) This species exhibits a reproductive characteristic known as sequential hermaphrodism. Which statement best describes the characteristic in this organism?

- A) All individuals are born male
- B) All individuals are born female
- C) All individuals are born with both sexes of reproductive organism and switch which one they use each year
- D) The sex of individuals at birth is random, but they will always transform into the other sex eventually

23. (1.00 pts) This species is demersal, which means they live only in brackish water

- True
- False

Refer to this image for questions 24-27



24. (1.00 pts) This organism belongs to a taxonomic class that is distributed in oceans around the globe

- True
- False

25. (1.00 pts) This organism is fit for human consumption

- (●) True
- ( ) False

26. (1.00 pts) Most species in this organism's class reproduce by external fertilization

- (●) True
- ( ) False

27. (1.00 pts) In 2020, India created a conservation area for these organisms. However, commercial harvesting and transportation of these organisms is allowed in the rest of India

- ( ) True
- (●) False

Refer to this image for questions 28-29



28. (1.00 pts) Why is this fish critical to the health of coral reefs?

- (●) A) They eat algae that would otherwise overgrow and smother the reef
- ( ) B) They prey upon invasive fish
- ( ) C) their bones are a significant source of calcium for organisms that develop calcium carbonate shells
- ( ) D) they form a commensal relationship with sea anemone

29. (1.00 pts) The IUCN considers this species to be a vulnerable species. Select all of the below factors that currently threaten this species

(Mark **ALL** correct answers)

- [x] A) overfishing
- [x] B) habitat degradation and destruction due to sediment run-off
- [ ] C) invasive species destroying their food sources
- [ ] D) high salinity

Refer to this image for questions 30-32



30. (1.00 pts) Identify the common name of this fish

Expected Answer: humphead wrasse

31. (1.00 pts) This fish is found in the Indo-Pacific region

(●) True ( ) False

32. (1.00 pts) The conservation of this species is important to other species because they are considered to be a(n)

(Mark **ALL** correct answers)

- [x] A) umbrella species
- [ ] B) indicator of good water quality
- [ ] C) important fish for aquariums
- [ ] D) very short-lived fish but with a long gestational period

Refer to this image for questions 33-35



33. (1.00 pts) Which statement best describes the reproductive habits of this organism? (Note: I'm not talking about the person)

- ( ) A) they use sexual reproduction with internal fertilization
- ( ) B) they are hermaphrodites that fertilize themselves
- (●) C) they are hermaphrodites that only fertilize each other

D) they are sequential hermaphrodites, with all individuals being born female

34. (1.00 pts) This species can only be found in the Atlantic region

True False

35. (1.00 pts) Complete the life cycle of this species: gametes to trocophores to _____ to pediveligers to juvilnes to adults

veligers

Refer to this image for questions 36-38



36. (1.00 pts) Identify the common name of this organism

Expected Answer: flamingo tongue snail

37. (1.00 pts) This species is only found in the Atlantic region

True False

38. (1.00 pts) Which best describes the reproductive habits of this species?

A) The males defend a cluster of eggs
B) The female uses an adhesive to attach eggs to coral
C) They use external fertilization and the eggs float with the currents until they hatch
D) They give live birth

Refer to this image for questions 39-40



39. (1.00 pts) This fish's common name refers to a capital city of Cuba

○ True ◉ False

40. (1.00 pts) This fish has a planktonic larval stage

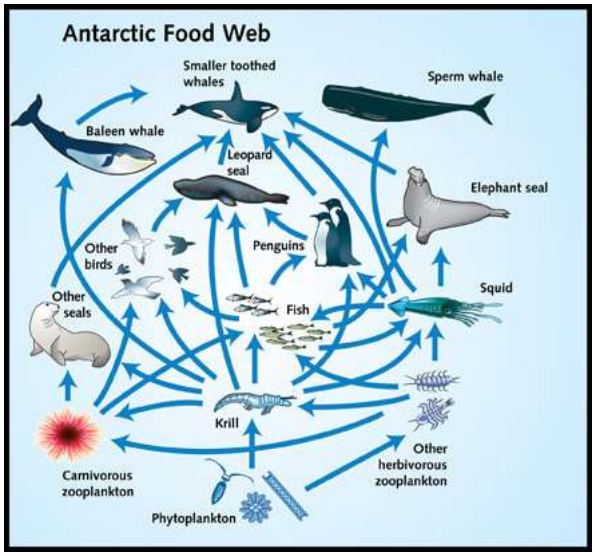
◉ True ○ False

Part 1: Marine and Estuary Ecology

41. (1.00 pts) Which term about interrelationships is best matched with its definition?

○ A) mutualism - beneficial for the host species
◉ B) amensalism - one species is negatively affected, one is neutral
○ C) commensalism - beneficial for both species

42. (1.00 pts) According to this food web, which of the following is a complete list of all the primary consumers?



- (●) A) krill - other herbivorous zooplankton
- ( ) B) krill - other herbivorous zooplankton - carnivorous zooplankton
- ( ) C) phytoplankton - other herbivorous zooplankton - carnivorous zooplankton
- ( ) D) baleen whale - smaller toothed whales - sperm whale

43. (1.00 pts) From the above food web, identify an autotroph

- (●) A) phytoplankton
- ( ) B) other herbivorous zooplankton
- ( ) C) carnivorous zooplankton
- ( ) D) krill

44. (1.00 pts) From the above food web, what would happen to the cephalopod population if the elephant seals were to disappear from the ecosystem?

- ( ) A) it would decrease
- (●) B) it would increase
- ( ) C) there is no relationship
- ( ) D) other seals would take their place in the food web and so it would stay the same

45. (1.00 pts) Which one of the following is considered a trophic level?

- ( ) A) the benthic layer
- ( ) B) troposphere
- ( ) C) subdermal
- (●) D) tertiary consumers

46. (1.00 pts) Which is the best definition for estuary?

- ( ) A) the intertidal zone
- ( ) B) a freshwater marsh
- (●) C) the area where the ocean meets a stream
- ( ) D) a place where marine birds are born

47. (1.00 pts) Phytoplankton in the open ocean are responsible for producing most of the planet's....

- (●) A) oxygen
- ( ) B) carbon
- ( ) C) ozone
- ( ) D) invasive species

48. (1.00 pts) What is the most likely place for a harmful algal bloom?

A) A lake in a mountainous area
B) The Mississippi River Delta
C) Upstream of a farm
D) a public swimming pool

49. (1.00 pts) Which of these statements best describes why the ocean can absorb a lot of thermal radiation, but not increase in temperature a lot?

A) Water does not conduct heat at all
B) Heat is absorbed by deep-ocean vents
C) Water has a relatively high specific heat
D) Water is non-polar, so its vibrational modes do not allow it to increase its kinetic energy quickly

50. (1.00 pts) What percentage of Earth's water is in the oceans?

A) 3
B) 50
C) 75
D) 97

51. (1.00 pts) What percentage of Earth's freshwater is found in the oceans?

A) 0
B) 3
C) 50
D) 97

52. (1.00 pts) What is the approximate salinity of the oceans?

A) 0-1%
B) 1-2%
C) 3-5%
D) 10%

53. (1.00 pts) Which compound is insoluble in water?

A) sodium chloride
B) potassium nitrate
C) ammonium chloride
D) calcium carbonate

54. (1.00 pts) Which of the following is not a consequence of biomagnification?

- ○ A) consumption of tuna should be limited
- ○ B) DDT was banned
- ◉ C) Lead in water can lead to brain damage in children
- ○ D) Polar compounds are found in high concentrations in predators

55. (1.00 pts) Which is not a form of chlorine used to disinfect water?

- ○ A) chlorine gas
- ◉ B) chloroform
- ○ C) sodium hypochlorite
- ○ D) calcium hypochlorite

56. (1.00 pts) What process is used in desalination plants to separate the salt and water?

- ◉ A) reverse osmosis
- ○ B) distillation
- ○ C) filtration
- ○ D) radiation

57. (1.00 pts) What is BOD?

- ○ A) A measure of nitrate ion concentration in water
- ○ B) A measure of the phosphate ion concentration in water
- ○ C) A measure of the amount of dissolved oxygen in water
- ◉ D) A measure of the amount of dissolved oxygen microorganisms use

58. (1.00 pts) Which type of water may be considered pure?

- ○ A) bottled spring water
- ○ B) filtered water
- ◉ C) distilled water
- ○ D) chlorinated water

59. (1.00 pts) Which of the following is responsible for coral reef bleaching?

- ○ A) The decreased solubility of aragonite (mineralized calcium carbonate)
- ◉ B) increased water temperatures
- ○ C) increased salinity
- ○ D) a lack of light for photosynthesis so the coral turns white

60. (1.00 pts) What symbol is used to represent carrying capacity?

- ◉ A) K

B) C
C) cc
D) delta

61. (1.00 pts) Which of the following lists has the 3 salinity zones for brackish water in the correct order from HIGHEST salinity to LOWEST?

A) polyhaline, mesohaline, oligohaline
B) oligohaline, mesohaline, polyhaline
C) oligohaline, polyhaline, mesohaline
D) polyhaline, oligohaline, mesohaline

62. (1.00 pts) What important species for human consumption has been depleted in the Chesapeake Bay and has led to a decrease in water quality?

A) sponges
B) oysters
C) phytoplankton
D) clams

63. (1.00 pts) Pharmaceuticals are impossible to remove at wastewater treatment plants

True False

64. (1.00 pts) Which of the following will not increase turbidity?

A) salt
B) sand
C) silt
D) clay

65. (1.00 pts) Select all greenhouses gases below

(Mark **ALL** correct answers)
A) carbon dioxide
B) water vapor
C) oxygen
D) methane

66. (1.00 pts) What is the primary form of sulfur taken up by organisms?

A) hydrogen sulfide
B) sulfur dioxide
C) sulfate
D) sulfite

67. (1.00 pts) What does PFAS stand for?

Expected Answer: per- and polyfluoroalkyl substances

68. (1.00 pts) Which of the following statements about PFAS contaminants is incorrect?

- A) PFAS chemicals are extremely persistent within our environment and bioaccumulate.
- B) PFAS chemicals can act as endocrine-disruptors due to their ability to interfere with hormone systems.
- C) Firefighting foams are a common source of PFAS chemicals.
- D) PFAS chemicals bind tightly to soils and organic matter, which removes them from water.

69. (1.00 pts) Acid mine drainage occurs due to the reaction of pyrite, FeS_2, with oxygen and water. What product causes the pH to decrease?

- A) $FeSO_4$
- B) H_2S
- C) $Fe(OH)_3$
- D) H_2SO_4

70. (1.00 pts) Where is most of the Earth's liquid freshwater found?

- A) groundwater
- B) rivers, ponds, lakes, and swamps
- C) springs
- D) ice caps

71. (1.00 pts) Which of the following is not a stage of the water cycle?

- A) evaporation
- B) percolation
- C) hydration
- D) transpiration

72. (1.00 pts) What is the best definition for a watershed?

- A) The land area that drains to a body of water
- B) The water stored in soil
- C) The water stored in a body of water
- D) A pumphouse used to treat and transport potable water

73. (1.00 pts) Which of the following are threats to watersheds? (select all)

(Mark **ALL** correct answers)
- [x] A) mining
- [x] B) farming
- [ ] C) overfishing
- [x] D) orbanization

74. (1.00 pts) The Great Pacific Garbage Patch can best be described as

- (●) A) A loosely dispersed but large collection of garbage
- ( ) B) A tightly packed patch of garbage that can be seen from space
- ( ) C) A collection of garbage that enters and exits the Pacific Ocean
- ( ) D) A designated zone in the Pacific Ocean for seagoing vessels to legally dump garbage

75. (1.00 pts) Which of these statements is incorrect regarding harmful algal blooms?

- (●) A) They can cause eutrophication
- ( ) B) They can cause a drastic depletion of oxygen in the water
- ( ) C) They can occur in both fresh and saltwater
- ( ) D) They can be caused by runoff from farms

76. (1.00 pts) The water that is treated in wastewater treatment plants is sent directly back to homes and businesses as it is now considered potable

( ) True (●) False

77. (1.00 pts) What are the three nutrients provided to crops with fertilizer?

- (●) A) nitrogen, phosporous, potassium
- ( ) B) nitrogen, potassium, magnesium
- ( ) C) nitrogen, phosphorous, calcium
- ( ) D) nitrogen, carbon, magnesium

78. (1.00 pts) The bottleneck effect is used to describe the common observance of marine organisms investigating a glass bottle but getting their head or limbs stuck inside

( ) True (●) False

79. (1.00 pts) Ocean acidification is primarily due to

- ( ) A) acidic runoff from farms
- ( ) B) acidic runoff from improperly installed septic tanks and improper design of wastewater treatment plants
- ( ) C) the anaerobic breakdown of garbage
- (●) D) carbon dioxide from the atmosphere dissolving into the ocean

80. (1.00 pts) According to the FDA "flush list" which of the following is okay to flush down the toilet?

(Mark **ALL** correct answers)

- [x] A) oxycodone
- [ ] B) anti-depressants
- [ ] C) birth control pills
- [ ] D) Bisphenol A

Part 3: Water Monitoring and Analysis

81. (1.00 pts) Untreated wastewater has a BOD of approximately _____ mg/L

- ○ A) 5.5
- ◉ B) 200
- ○ C) 10
- ○ D) 1000

82. (1.00 pts) Which of the following is not a source of nitrate contamination of water sources?

- ○ A) agricultural practices
- ○ B) rock weathering
- ○ C) septic systems
- ◉ D) pipe corrosion

83. (1.00 pts) Excessive _____ causes hard water

- ○ A) phosphate
- ○ B) fluoride
- ◉ C) calcium
- ○ D) freezing

84. (1.00 pts) What does low dissolved oxygen levels indicate in water bodies?

- ○ A) The water is highly acidic caused by acid rain or industrial waste contamination.
- ◉ B) The water quality is low due to high levels of bacteria from sewage pollution or large amounts of decomposing plants.
- ○ C) Limestone rock was weathered in the water body, causing alkaline water.
- ○ D) The natural water body includes plants and fish that are consuming the oxygen.

85. (1.00 pts) How is the pH of a water sample most precisely measured?

- ◉ A) potentiometry
- ○ B) spectrophotometry
- ○ C) colorimetric analysis

D) ascorbic acid method

86. (1.00 pts) What can happen to the pH of a water body that has limestone nearby?

A) the pH could be neutralized
B) Decreases the pH
C) Increases the pH
D) The pH of the water body won't be affected

87. (1.00 pts) Select all of the misconceptions (incorrect statements) about pH below

(Mark **ALL** correct answers)
A) It measures the concentration of hydrogen (or hydronium) ions in solution
B) the lower the pH, the more corrosive the acid is
C) it is a log scale, so small changes in pH can represent large differences in the concentration of acid
D) the lower the pH, the stronger the acid is

88. (1.00 pts)
A wastewater treatment facility wanted to measure the BOD of untreated wastewater. A 10.0 mL sample was added to 150. mL of water. The initial dissolved oxygen concentration was 7.5 mg/L. After 5 days the dissolved oxygen concentration was 1.4 mg/L. What is the BOD of this sample, measured in mg/L?

A) 91
B) 0.038
C) 130
D) 0.055

89. (1.00 pts) Which of the following methods can remove turbidity in water?

A) aeration
B) acidification
C) UV sanitation
D) coagulation

90. (1.00 pts) What device is commonly used to measure turbidity?

A) pH meter
B) secchi disk
C) salinometer
D) conductivity meter

91. (1.00 pts) What would happen if concentrated waste CO_2 was discarded 5000 meters below the surface of the ocean?

A) CO_2 would dissolve in water causing the pH to increase

B) CO_2 would dissolve in water causing the pH to decrease
C) CO_2 would be trapped in an ice-like structure causing no change in pH
D) CO_2 would be trapped in an ice-like structure, pulling more CO_2 into it, causing the pH to increase

92. (1.00 pts) Total dissolved solids in water include

A) gases
B) colloids
C) sediment
D) minerals

93. (1.00 pts) The concentration of dissolved solids in water can be determined most easily by

A) specific conductance
B) weight of residue after evaporation
C) the sum of concentrations determined by chemical analysis
D) specific gravity determination

94. (1.00 pts) How does salt end up in the oceans?

A) salt was always present in the oceans
B) volcanoes let out salt when they erupt
C) the water that washed over rocks carried salt into the oceans
D) from acid rain that falls into the ocean

95. (1.00 pts) The salinity in water is measured in ppt. What does ppt in this context stand for?

A) parts per tens
B) parts per thousand
C) particles pertaining titanium
D) precipitation

96. (1.00 pts)
Water is able to exist on Earth as a liquid instead of only a gas or solid is due to its relatively high boiling point. What is the reason water has a relatively high boiling point?

A) It has a high specific heat
B) It has high attractive forces between water molecules
C) Water has a high vapor pressure
D) It expands as it freezes

97. (1.00 pts) What allows a salinometer to be calibrated and work?

A) the density of salt water at varying concentrations is predictable

B) the specific heat of salt water at varying concentrations is predictable
C) the conductivity at varying concentrations is predictable
D) the temperature of of salt water at varying concentrations is predictable

98. (1.00 pts) A water sample analysis has shown a homeowner that their well is contaminated with fecal coliforms. What is not a proper technique for disinfecting the water?

A) boiling the water for several minutes
B) treating with chlorine
C) UV disinfection
D) treating with acidic agents and then neutralizing

99. (1.00 pts) Fecal coliform do not require oxygen to decompose organic matter

True False

100. (1.00 pts) As temperature increases, the BOD _____ and dissolved oxygen tends to _____

A) increase, increase
B) increase, decrease
C) decrease, decrease
D) decrease, increase

101. (1.00 pts) A pH of 5 has _____ times more dissolved acid concentration than a sample of water with a pH of 7

A) 2
B) 10
C) 100
D) 5/7

102. (1.00 pts) dead zones are areas where no organisms can be sustained because they are best described as

A) dehydrated
B) alkaline
C) hypoxic
D) too hot

103. (1.00 pts)
A point source contributes sulfur dioxide and nitrogen oxides into the atmosphere. A body of water downwind would over time have a pH that is _____ than beforehand

A) the same
B) higher
C) lower
D) there is not enough information to determine

104. (1.00 pts) An example of a disease associated with high levels of fecal coliform in drinking water is

- A) influenza
- B) hepatitis A
- C) measles
- D) arthritis

The figure below is a global map of saturation of aragonite. The pink dots are locations of coral reefs.

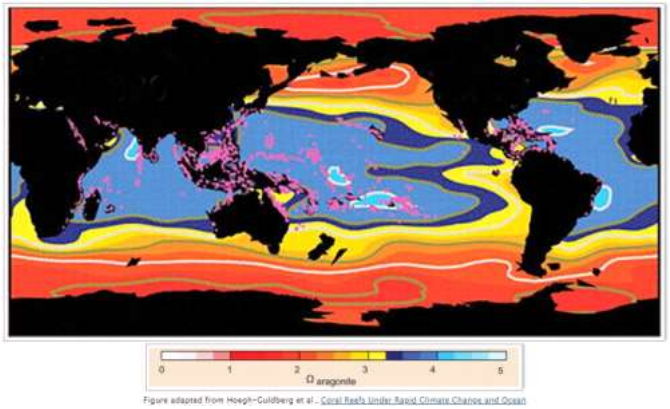


Figure adapted from Hoegh-Guldberg et al., Coral Reefs Under Rapid Climate Change and Ocean

105. (1.00 pts) On this map, what does the Greek letter omega represent?

- A) The calcium carbonate saturation state
- B) The total calcium carbonate dissolved
- C) The population of coral in the oceans
- D) The relative ocean temperature

106. (1.00 pts) What would happen to the value of omega if more carbon dioxide was dissolved in the ocean?

- A) decrease
- B) increase
- C) stay the same
- D) there is not enough information to determine

107. (1.00 pts) Which ion does not contribute to alkalinity?

- A) bicarbonate
- B) carbonate
- C) hydronium
- D) hydroxide

108. (1.00 pts) Alkalinity helps protect aquatic ecosystems from dramatic changes in

- (●) A) pH
- ( ) B) temperature
- ( ) C) nitrate concentration
- ( ) D) harmful algae population

109. (1.00 pts) The amount of phosphate in a water body is usually measured in

- ( ) A) ppt
- (●) B) ppm
- ( ) C) ppb
- ( ) D) percent

110. (1.00 pts) Why should dissolved oxygen be measured right at the water body of interest?

- (●) A) if the temperature gets too warm, the oxygen will become less soluble
- ( ) B) if the water sloshes around too much, the reading will be unreliable
- ( ) C) If the temperature gets too cold, the oxygen will become less soluble
- ( ) D) zooplankton in the water will continue to produce oxygen during the transportation of the water sample

111. (1.00 pts) The WHO suggests drinking water should not be above ____ NTU

- ( ) A) 2
- ( ) B) 5
- (●) C) 10
- ( ) D) 100

112. (1.00 pts) Under normal circumstances, one could expect the amount of dissolved oxygen at night to

- ( ) A) be higher than during the day
- (●) B) be lower than during the day
- ( ) C) be about the same as during the day
- ( ) D) there is not enough information to determine

113. (1.00 pts) What is the highest possible value for albedo?

- ( ) A) 10
- ( ) B) 14
- (●) C) 1
- ( ) D) There is no limit

114. (1.00 pts) The albedo of the ocean is _____ the albedo of bare sea ice

A) about the same as
B) much less than
C) much higher than
D) twice as much as

115. (1.00 pts) TSS can reduce photosynthesis by

A) disrupting photosystem 1
B) it is a toxin specific to phytoplankton and cyanobacteria
C) it physically blocks the sunlight from reaching zooplankton
D) it physically blocks the sunlight from reaching autotrophs

116. (1.00 pts) Overuse of road salt on roads during the winter can lead to an increase in ____ in local bodies of water

A) TDS
B) TSS
C) BOD
D) DO

117. (1.00 pts) Nitrogen makes up almost four-fifths of the atmosphere. Why does it still need to be given to crops in fertilizer?

A) Nitrogen is too large to diffuse across the membrane of plant cells
B) Crops are unable to fix nitrogen
C) The crops are genetically modified and need to be sensitized to nitrogen first
D) The partial pressure of nitrogen in the atmosphere is still too low

118. (1.00 pts) What type of organism would survive the best in a high salinity environment?

A) acidophile
B) thermophile
C) halophile
D) sodiophile

119. (1.00 pts) What type of organism would survive the best in an environment with that fluctuates between normal and almost zero DO?

A) Obligate aerobe
B) Obligate anaerobe
C) Facultative anaerobe
D) No organism can survive this fluctuation

120. (1.00 pts) Phosphate runoff is common from fertilizer used on farms and is important to monitor. What mineral is a source of phosphate used in fertilizer?

A) halite

- (●) B) apatite
- ( ) C) pyrite
- ( ) D) aragonite

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